

GEN AI Bootcamp

Problem Statement

Designing an AI system for an e-commerce platform (like Amazon or Flipkart) that can understand a product using its image — and produce a category and description automatically.

1. Inputs

- **Product image** — uploaded by the seller at listing time.

2. Outputs

- **Product category** (e.g., "Footwear > Sneakers")
- **Product description** (e.g., "White canvas sneakers with rubber sole, lace-up closure, casual style")

3. Model Selection

The task needs a **vision-language model (VLM)** — one that understands both images and text. Two complementary model types fit the two outputs:

- **CLIP** (Contrastive Language-Image Pretraining) — learns a shared embedding space for images and text. Ideal for **classification/retrieval**: compare the product image embedding against category-label embeddings to find the best match. It does not generate free-form text.
- **BLIP / BLIP-2** — an image-captioning model that **generates** natural language directly from an image.

Best suited for producing the description. • **Newer alternative**: a single large multimodal model (GPT-4V-class or LLaVA-style VLM) can do both classification and description generation in one pass with a single prompt.

Chosen approach: CLIP for category classification (fast, cheap, works well with a fixed label set) + BLIP/BLIP-2 for generating the description.

4. System Design (High-Level)

Step-by-step flow of the system, from image upload to the data being used downstream:

Multimodal Product Understanding — System Design

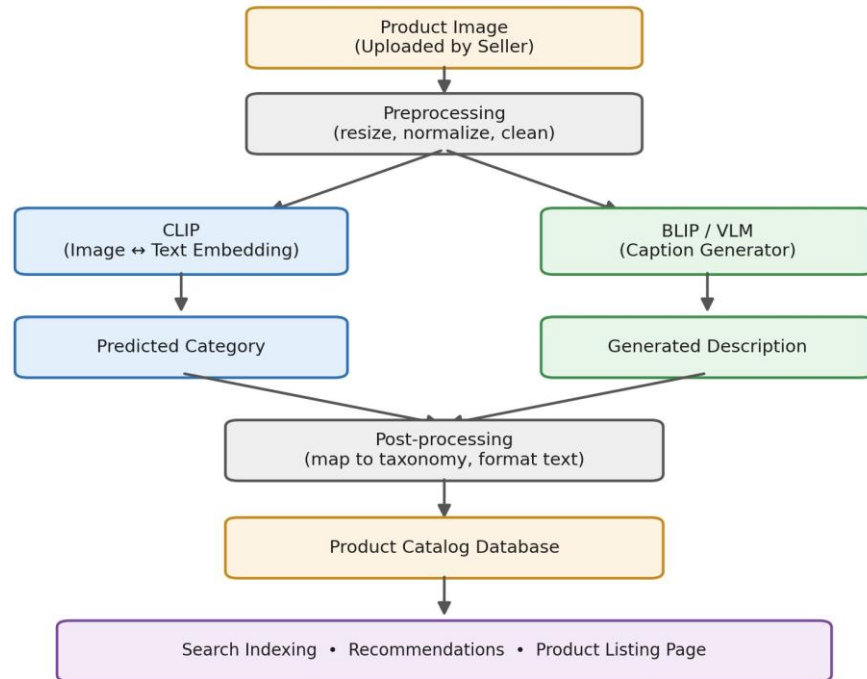


Figure 1: High-level multimodal product understanding pipeline

- 1. Seller uploads a product image. •
- 2. Image is preprocessed (resized, normalized).
- 3. CLIP embeds the image and matches it against pre-defined category embeddings → predicted category. •
- 4. BLIP (or a VLM) generates a caption/description from the same image. •
- 5. Both outputs are cleaned up and mapped to the platform's taxonomy/formatting rules. •
- 6. Results are saved to the catalog and surfaced in listings, search indexing, and recommendations.

5. Evaluation

Category accuracy:

- Top-1 / Top-3 accuracy, F1-score per category.
- Confusion matrix — catches categories the model frequently mixes up.

Description quality:

- Automated: BLEU / ROUGE / METEOR against human-written reference descriptions, or **CLIPScore** (measures how well generated text matches the image semantically — useful when reference text isn't available). •

Human evaluation: sample reviews for fluency, relevance, and whether key attributes (color, material, style) are correctly captured.

Business-facing metric:

- % of listings that needed manual correction after auto-generation — a strong proxy for real-world usefulness.

6. Business Value

- **Faster seller onboarding** — sellers just upload a photo instead of writing full listings. •
- **Catalog consistency** — uniform category tagging improves search and filtering. • **Better discovery** — accurate categories/descriptions improve search relevance and recommendation quality. •
- **Cost savings** — reduces manual cataloging labor at scale (huge for platforms with millions of SKUs). •
- **Accessibility / SEO** — auto-generated descriptions improve product page text for search engines and screen readers.

Bonus: Real-World Example

Pinterest uses multimodal AI in its visual search feature ("Pinterest Lens"): a user points their camera or uploads a photo of an object, and Pinterest matches it against visually and semantically similar pins/products using image-text embeddings, then surfaces shoppable results. This mirrors the CLIP-based matching step used in the system design above.